Cover Letter

November 13, 2026

Lecturer Search Committee
Elmore Family School of Electrical and Computer Engineering
Purdue University in Indianapolis
Indianapolis, Indiana

Re: Lecturer, Purdue Indianapolis – Electrical & Computer Engineering (Req. 31383)

Dear Members of the Search Committee:

I am applying for the full-time Lecturer position in Electrical and Computer Engineering at Purdue University in Indianapolis. I earned my PhD in Electrical and Computer Engineering from the University of Oklahoma in 2025, specializing in AI and machine learning for reliable wireless networks. My background combines undergraduate university teaching, engineering laboratory instruction, student research mentoring, and applied engineering experience at AT&T, AT&T Labs, Futurewei, and Impac Exploration Services. I am particularly interested in contributing to Purdue's undergraduate instruction, course development, and student-centered engineering education.

As a Lecturer at the University of Lahore, I taught Computer Programming, Communication Systems, and Wireless Communication, prepared course materials and assessments, evaluated student learning, and guided projects. I also supported communications laboratories and assessment at the University of Engineering and Technology, Taxila, and mentored research students at COMSATS and the University of Oklahoma. These experiences have shaped my approach to instruction: connecting analytical foundations with implementation, experimental evaluation, and clear feedback. My strongest teaching areas include programming, communication systems, wireless networks, signals and systems, digital signal processing, and applied AI/ML for engineering. I would prepare additional lecture and laboratory assignments in consultation with faculty and the approved curriculum.

My doctoral and industrial work provides practical examples that can enrich undergraduate engineering instruction. In my current machine-learning engineering assignment at AT&T, I develop and evaluate AI-based diagnostics for E911 service assurance; one root-cause classification workflow improved automated classification performance from approximately 20% to 80%. My experience with Python, MATLAB, C/C++, cloud analytics, and reproducible experimentation can support computational laboratories, technical projects, and responsible use of modern AI tools. Purdue Indianapolis's activities in autonomous systems and intelligent software offer particularly relevant opportunities for interdisciplinary engineering examples.

I recognize that this position entails primary responsibility for three courses each fall and spring, including assessment, office hours, course improvement, and supervision of teaching assistants. I would approach these responsibilities through organized course schedules, shared grading rubrics, laboratory preparation, regular TA coordination, and timely communication. I would use Microsoft Word, Excel, Teams, and PowerPoint alongside appropriate programming and learning tools to support consistent instructional delivery. My teaching statement describes how I would integrate AI-assisted engineering work while preserving independent reasoning and assessment integrity.

I would welcome the opportunity to contribute to Purdue's electrical and computer engineering curriculum and collaborate with colleagues on effective undergraduate education. Thank you for considering my application. I look forward to discussing how my teaching and industry experience can support your students.

Sincerely,

Waseem Raza, PhD